University of Regina – Department of Biology
Biology 835AQ – Data Science in Ecology
May 2026 - August 2026

Instructor
Bruno Soares
bruno.soares@uregina.ca
Boardroom, Institute of Environmental Change and Society

Description

A Directed Readings course for graduate students facilitating them to explore best practices in reproducible ecological data science using R. Students will have one-on-one meetings, literature engagement, and guided demonstrations on reproducibility, metadata, tidy workflows, project organization, and transparent analysis. Through the course, they will develop an independent ecological data pipeline, either methodological (e.g., building functions or workflows) or ecological (addressing a research question). The final product must demonstrate ecological insight, good workflow design, and reproducible practices. This course supports the development of key skills to use available tools for data management and analysis.

In addition to the core seminar meetings, the course includes regular data studio sessions. These sessions focus on hands-on work with student projects, troubleshooting workflows, and applying reproducible data science practices in real time.

Course Structure

Phase 1 – Project Kickoff (June 29 – July 10)
Phase 2 – Skill Building and Project Start (July 13 – July 24)
Phase 3 – Pipeline Development (Late July)
Phase 4 – Refinement and Integration (Early August)
Phase 5 – Final Presentations (Mid-Late August)

Grade Allocation

Component	**Due date**	**Weight**	**Description**
Ignite Presentation	July 10	10%	Up to 5-minute presentation describing project motivation, relevance, expected structure, and goals. You should schedule this presentation with the instructor.
Pipeline - Draft 1	July 31	20%	Early version of workflow: folder structure, metadata, data import + cleaning steps, initial code annotations, and at least one preliminary output. This should be delivered as a GitHub repository.
Workshop Participation	-----	15%	Participating at the provided workshops in July.
Final Presentation	August 27	15%	5-minute presentation describing project motivation, relevance, achieved structure, and challenges. You should schedule this presentation with the instructor.

Final Pipeline + Manuscript	August 21	40%	Final cleaned and annotated pipeline (or minimal R package/workflow) with metadata, README, and a short (up to 5 pages) manuscript explaining ecological insight, workflow reasoning, results demonstration, and limitations. This should be delivered as a GitHub repository containing clean and annotated code, metadata files, instructions, and the manuscript.

Project Types

Methodological Project: students develop a workflow, pipeline, or set of functions addressing a methodological challenge, such as calculating ecological indices, downloading and cleaning phylogenies. wrangling trait data, cleaning biodiversity occurrences, developing API-based data processing scripts. Here, the focus will be reproducibility, implementation, metadata, tool clarity, and how the work fills a potential gap in available tools.

Ecological Question Project: students analyze a dataset to answer an ecological question using a reproducible workflow. Focus: ecological reasoning, transparency and adequacy of choices, and clear, reproducible pipeline structure.

Data Rescue Project Rescue: students reorganize, document, and validate a dataset (or integrated dataset) and produce a manuscript-style data paper describing data provenance, structure, quality control, and reuse potential.

Readings and Supporting Materials

Borer, E.T., Seabloom, E.W., Jones, M.B., Schildhauer, M. (2009). Some simple guidelines for effective data management. Bulletin of the Ecological Society of America, 90, 205-214.

British Ecological Society. A guide to data management in Ecology and Evolution.

Broman, K.W., Woo, K.H. (2018). Data organization in spreadsheets. American Statistician, 72, 2-10.

Carroll, M.W. (2015). Sharing research data and intellectual property law: a primer. PLoS Biology, 13, e1002235.

de Jonge, E., van der Loo, M. (2013). An introduction to data cleaning with R. Statistics Netherlands.

Gould, E., Fraser, H.S., Parker, T.H. et al. (2025). Same data, different analysts: variation in effect sizes due to analytical decisions in ecology and evolutionary biology. BMC Biol 23, 35.

Harrison XA, Donaldson L, Correa-Cano ME, Evans J, Fisher DN, Goodwin CED, Robinson BS, Hodgson DJ, Inger R. 2018. A brief introduction to mixed effects modelling and multi-model inference in ecology. *PeerJ* 6:e4794

Popovic, G., Mason, T. J., Drobniak, S. M., Marques, T. A., Potts, J., Joo, R., Altwegg, R., Burns, C. C. I., McCarthy, M. A., Johnston, A., Nakagawa, S., McMillan, L., Devarajan, K., Taggart, P. L., Wunderlich, A., Mair, M. M., Martínez-Lanfranco, J. A., Lagisz, M., & Pottier, P. (2024). Four principles for improved statistical ecology. Methods in Ecology and Evolution, 15, 266–281.

Roche, D.G., Kruuk, L.E., Landfear, R., Binning, S.A. (2015). Public data archiving in ecology and evolution: how well are we doing? *PLoS Biology*, 13, e1002295.

William, M., Bagwell, J., Nahm Zozuz, M. (2017). Data management plans: the missing perspective. *Journal of Biomedical Informatics*, 71, 130-142.

Plagiarism
Plagiarism is using the words, ideas or thoughts of others and presenting them as your own. All students must read "Recognizing and Avoiding Plagiarism" located here:
https://www.uregina.ca/student/ssc/assets/docs/pdf/Recognizing-and-Avoiding-Plagiarism.pdf
Students must also review the Plagiarism Spectrum located here:
https://www.turnitin.com/static/plagiarism-spectrum/

Accommodations
Students in this course who may require specialized accommodations should contact the Center for Student Accessibility (https://www.uregina.ca/student/accessibility/). Accommodations must be requested each term, and are not retroactive.